Code No: 152AF **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
PROGRAMMING FOR PROBLEM SOLVING
(Common to EEE, CSE, IT, ITE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Differentiate between typecasting and type conversion.
b) Write a program to count the number of vowels in a text. [7+8]

2.a) Write a program to read two numbers. Then find out whether the first is a multiple of the second number.
b) Explain the use of Command line argument with an example. [8+7]

3.a) Describe about the enumerated data type with an example.
b) Explain about the use of pointers in self referential structures. [7+8]

4.a) Write a program to transpose of a 3×3 matrix.
b) Differentiate between structure and an array. [8+7]

5.a) Explain the importance of #define preprocessor directive.
b) Write a program to read a file that contains lowercase characters. Then write these characters into another file with lower case characters converted into upper case. [7+8]

6.a) Write a program to merge two files into a third file. The names of the files must be entered using command line arguments.
b) Distinguish between #ifdef and #ifndef. [8+7]

7.a) Write a recursive program to find out the factorial of given number.
b) Explain about the dynamic memory allocation. [8+7]

8.a) Write an algorithm to find out the minimum and maximum numbers of a given set.
b) Explain about the Bubble sort with an example. [7+8]

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